AQ8.1: Activity Questions 1 - Not Graded

This assignment will not be graded and is only for practice.
Level 1

1 point
Consider two vectors a = (2, 0, 3, 0, 8), b = (3, 2, − 2, 4, 0)a = (2, 0, 3, 0, 8), b = (3, 2, −2, 4, 0) in R^5 R5.
Choose the set of correct options.

- [ ] aa and bb are orthogonal.
- [ ] aa and bb are not orthogonal.
- [ ] $(a - b) \cdot a$ = 0.(a − b) · a = 0.
- [ ] $(a - b) \cdot a$ = 77.(a − b) · a = 77.

No, the answer is incorrect.
Score: 0
Accepted Answers:
aa and bb are orthogonal.
$(a - b) \cdot a$ = 77.(a − b) · a = 77.

1 point
Choose the set of correct statements.

- [ ] In an orthogonal set, the norms of all the vectors are equal.
- [ ] In an orthogonal set, the vectors are linearly independent.
- [ ] In an orthogonal set, the vectors are linearly dependent.
- [ ] If the columns of an $n \times n$n × n coefficient matrix AA comprises of the individual vectors of an orthogonal set in R^n Rn, then there must be a unique solution to the system $AX = b$AX = b, where X, bX, b are $n \times 1$n × 1 vectors.
- [ ] If the columns of an $n \times n$n × n coefficient matrix AA comprises of the individual vectors of an orthogonal set in R^n Rn, then there are no solutions to the system $AX = b$AX = b, where X, bX, b are $n \times 1$n × 1 vectors.
- [ ] The determinant of a square matrix formed by a set of orthogonal vectors in R^n Rn is zero.
- [ ] A set of nn vectors can never form an orthogonal basis in R^{n-1}.Rn−1.

No, the answer is incorrect.
Score: 0
Accepted Answers:
In an orthogonal set, the vectors are linearly independent.
If the columns of an $n \times n$n × n coefficient matrix AA comprises of the individual vectors of an orthogonal set in R^n Rn, then there must be a unique solution to the system $AX = b$AX = b, where X, bX, b are $n \times 1$n × 1 vectors.
A set of nn vectors can never form an orthogonal basis in R^{n-1}.Rn−1.

1 point
Which of the following is an orthogonal basis of the given vector spaces with respect to the sta
inner product (dot product)?

- [ ]

$\{(1,0),(0,1)\}\{(1,0),(0,1)\}$ is an orthogonal basis of R^2 R2.

$\{(1,0,0),(0,1,0),(0,0,1)\}\{(1,0,0),(0,1,0),(0,0,1)\}$ is an orthogonal basis of R^3 R3.

$\{(3,4),(4,\ -3),(2,\ -3)\}\{(3,4),(4,-3),(2,-3)\}$ is an orthogonal basis of R^2 R2.

$\{(2,1,\ -1),(\ -1,1,\ -1),(3,\ -3,3)\}\{(2,1,-1),(-1,1,-1),(3,-3,3)\}$ is an orthogonal basis of R^3 R3.

No, the answer is incorrect.
Score: 0
Accepted Answers:
$\{(1,0),(0,1)\}\{(1,0),(0,1)\}$ is an orthogonal basis of R^2 R2.
$\{(1,0,0),(0,1,0),(0,0,1)\}\{(1,0,0),(0,1,0),(0,0,1)\}$ is an orthogonal basis of R^3 R3.

Level 2

1 point
Find a vector in R^4 R4 that is orthogonal to the subspace spanned by $(1,1,0,0)(1,1,0,0)$ and $(0,1,1,0)$ $(0,1,1,0)$ with respect to the dot product as the inner product.

- (1,-1,1,0)
- (2,3,4,5)
- (1,-1,1,1)
- (1,1,1,0)

No, the answer is incorrect.
Score: 0
Accepted Answers:
(1,-1,1,0)
(1,-1,1,1)

1 point
Which of the following are orthogonal to the vector $(1,2)(1,2)$ in R^2 R2 with respect to the inner product $<v,w>\ = v_1 w_1 - v_1 w_2 - v_2 w_1 + 4v_2 w_2$ < v, w >= v1w1 − v1w2 − v2w1 + 4v2w2?

$(-7,1)(-7,1)$

$(6,\ -1)(6,-1)$

$(7,1)(7,1)$

$(-9,1)(-9,1)$

$(14,2)(14,2)$

No, the answer is incorrect.
Score: 0
Accepted Answers:
$(7,1)(7,1)$
$(14,2)(14,2)$

Consider the system of linear equations:

$x_1 - 2x_2 + 3x_3 = 1$x1 − 2x2 + 3x3 = 1

$2x_1 + x_2 = 52x1 + x2 = 5$

$3x_1 - 6x_2 - 5x_3 = 9.3x1 - 6x2 - 5x3 = 9.$

Number of solutions of the above system is

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 1
1 point
1 point
Let AA be the coefficient matrix of the system of linear equations:

$x_1 - 2x_2 + 3x_3 = 1x1 - 2x2 + 3x3 = 1$

$2x_1 + x_2 = 52x1 + x2 = 5$

$3x_1 - 6x_2 - 5x_3 = 9.3x1 - 6x2 - 5x3 = 9.$

Which one of the following is true about the matrix AA^T AAT ?
- [ ] A scalar matrix.
- [ ] The identity matrix.
- [ ] A diagonal matrix.
- [ ] A lower triangular matrix.
- [ ] An upper triangular matrix.
- [ ] None of the above.

No, the answer is incorrect.
Score: 0
Accepted Answers:
A diagonal matrix.
A lower triangular matrix.
An upper triangular matrix.

Check Answers
Your score is: 0/7